

Mathana Mathav A.S.

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EXPERIENCE

- **DataGenie** Remote
Founding Engineer, Data & AI December 2023 – Present
 - Optimize production PySpark jobs processing **10M+ records** daily at **99.9% uptime**, including running anomaly detection on aggregated data losslessly compressed to a **third** of its original size.
 - Design and maintain core microservices built from the ground up for metrics, insights, pipeline orchestration and LLM integration, cutting metric query latency from **2–3s to under 500ms**.
 - Implement analytics algorithms such as contribution analysis and Shapley value attribution to explain which segments drove a change in a KPI.
 - Work on DataGenie's agentic BI chat for business users, across the agent harness, guardrails, query planning, and the visualization layer that turns answers into shareable reports.
 - Work directly with customers to understand business pain points, translate them into product requirements, and drive adoption of the features we ship while maintaining customer relationships.
- **ZeroDown (YC W19)** Remote
Software Engineer Intern June 2022 – December 2022
 - Maintained existing ETL pipelines that enriched **1M+ US home listings** with neighborhood data such as nearby shops, and built automated ETL pipelines to integrate new data sources.
 - Designed an automated incident routing system that analyzes error tracebacks to route **~85% of errors** to the responsible developer, cutting the time spent on initial debugging.

PROJECTS

- **IntersectX**: Built an LLM chat and investment research platform where founders pitch their companies and VCs research them; a central MCP orchestrator coordinates specialized agents (finance, market, team, risk) backed by vector search and live web research. (Demo | GitHub)
- **Guardian Agents**: Built a women's safety companion app where AI agents assess risk in real time from location safety scores, traffic and past incident data, recommend the safest route, and trigger emergency alerts. (GitHub)
- **DoodleDraw**: Trained a CNN to recognize doodles in real time for a browser drawing game that scores players when their sketch matches the prompt. (GitHub)

SKILLS

- **Languages & Frameworks**: Python, FastAPI, Flask, Django, Streamlit
- **AI Frameworks**: Agno, LangChain, FastMCP, TensorFlow
- **Data**: PySpark, Airflow, MongoDB, PostgreSQL
- **Cloud & DevOps**: Azure, GCP, Docker, Kubernetes

EDUCATION

- **PSG College of Technology** Coimbatore, India
Integrated M.Sc. Data Science; CGPA 8.67 June 2019 – April 2024
- **Adhyapana CBSE School** Madurai, India
Class XII (CBSE); Marks 93% 2018 – 2019

ACHIEVEMENTS

- **Code For Her**: Won first prize at Locus's Code For Her hackathon, 2025 (Article | LinkedIn).
- **IntersectX**: Prize winner at Agno's Global Agent Hackathon, 2025, among 60+ final submissions (Winners).
- **Open source**: Bunker, an attendance tracking app, is still used by some students at PSG (GitHub).